

AdaGeoVLN: Hierarchical Geometry Fusion and Navigation-Aware Memory for Vision-Language Navigation

Anonymous Authors

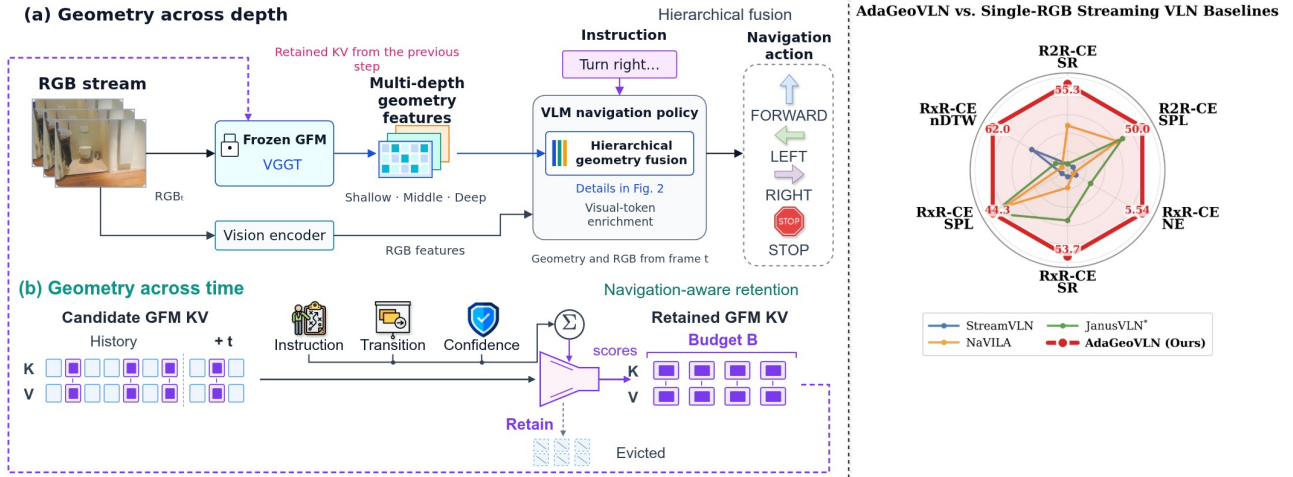


Fig. 1: **AdaGeoVLN overview.** Geometry is selected along two complementary axes. Across *representation depth*, intermediate VGGT states are coupled to successive VLM stages. Across *navigation time*, historical VGGT global-attention KV states are selectively retained under a bounded memory budget. The right panel summarizes performance relative to representative single-RGB streaming VLN baselines on R2R-CE and RxR-CE Val-Unseen.

Abstract—Vision-language navigation requires an embodied agent to align language with visual observations while maintaining a spatially coherent understanding of an environment over time. Geometry foundation models (GFM) expose intermediate representations throughout their processing hierarchy, yet it remains unclear whether VLN should rely on a single terminal geometric representation or interact with geometry across multiple representational depths. Long-horizon navigation introduces a second challenge: geometric evidence accumulates over time, but not all historical states remain equally useful for future decisions. We introduce AdaGeoVLN, a streaming VLN framework that addresses these two selection problems jointly across representation depth and navigation time. First, we propose hierarchical GFM–VLM fusion, which couples earlier, intermediate, and later GFM representations to successive stages of the navigation policy instead of repeatedly reusing a single terminal feature. Second, we introduce navigation-aware GFM memory, which retains historical VGGT global-attention KV states according to instruction relevance, geometric confidence, and transition novelty under a bounded per-layer memory budget. Across R2R-CE and RxR-CE, AdaGeoVLN achieves strong performance in the single-RGB streaming setting without relying on additional navigation-specific external data. Controlled ablations show that multi-depth geometric coupling substantially outperforms repeated terminal-feature injection, while bounded navigation-aware retention preserves navigation performance while considerably reducing GFM memory usage. These results suggest that effective geometric reasoning for embodied navigation depends not only on introducing 3D information, but on selectively allocating geometric representations across both model depth and navigation history.

I. INTRODUCTION

Vision-language navigation (VLN) requires an embodied agent to interpret a natural-language instruction, perceive an unfamiliar environment, and execute a sequence of actions that remains consistent with both the instruction and the evolving spatial context. Recent vision-language models (VLMs) have substantially improved semantic understanding and instruction following, enabling navigation policies to reason directly from RGB observations and language. However, successful navigation is fundamentally geometric: an agent must distinguish traversable structure, reason about viewpoint changes, maintain spatial relationships across observations, and recognize when previously observed geometry remains relevant to a future decision. These requirements become increasingly important in continuous and long-horizon navigation, where the same semantic scene can induce different actions depending on the agent’s position, orientation, and navigation history.

Geometry foundation models (GFM) offer a promising source of such spatial information. Models such as VGGT [1] learn geometric representations from large-scale visual data and infer scene structure without requiring task-specific depth or pose supervision at deployment. Nevertheless, *how geometric representations from a GFM should interact with a language-conditioned navigation policy remains largely unresolved*. A common design choice is to expose the policy

to a single geometry representation, often taken from the final stage of a geometry encoder. This implicitly assumes that a terminal representation alone contains the geometric information most useful for navigation. Yet a deep GFM exposes intermediate representations throughout its processing hierarchy, raising a more fundamental question: *should navigation reasoning consume geometry only after it has been collapsed into a terminal representation, or should geometric information from multiple representation depths interact with different stages of policy reasoning?*

This question is distinct from simply injecting more geometric features. Repeatedly introducing the same terminal representation at several locations increases the amount of geometry-policy interaction but does not introduce information from different points of the GFM hierarchy. We therefore study **geometry selection across representation depth**. Our hypothesis is not that particular GFM layers correspond to predefined semantic categories, but that intermediate representations may contain complementary information that is lost when geometric reasoning is reduced to a single terminal feature. Based on this hypothesis, we introduce a hierarchical GFM-VLM coupling mechanism in which representations from different GFM depths are aligned with successive stages of the navigation policy. This formulation allows us to test explicitly whether the benefit arises from multi-depth geometric representations rather than from repeated geometry injection alone.

A second and orthogonal problem emerges over **navigation time**. Unlike static visual reasoning, navigation produces a continually growing stream of observations. A geometry model can accumulate historical states to preserve spatial information across viewpoints, but unrestricted retention causes memory to grow with trajectory length. Conversely, retaining states solely according to recency assumes that recently observed geometry is always more useful than older geometry, which need not hold in instruction-conditioned navigation. An observation encountered many steps earlier may remain important because it corresponds to an instruction landmark, a reliable geometric observation, or a significant transition in the trajectory. This motivates a second scientific question: *which historical geometric states should remain available to future geometric reasoning?*

We address these two questions jointly with AdaGeoVLN, a framework for selective geometric reasoning along two orthogonal axes: **representation depth** and **navigation time**. Across representation depth, AdaGeoVLN hierarchically couples earlier, intermediate, and later GFM representations with successive VLM reasoning stages. Across time, it maintains a bounded VGGT global-attention memory and selects historical states according to navigation utility, combining instruction relevance, geometric confidence, and transition information. Importantly, the temporal mechanism operates within the geometry model itself: retained GFM states influence future geometric reasoning before geometry is subsequently fused into the navigation policy. Fig. 1 summarizes this view of AdaGeoVLN as selective geometry allocation across depth and time.

We evaluate this hypothesis-driven design through both benchmark-scale evaluation and controlled analysis. On the standard R2R-CE and RxR-CE benchmarks, AdaGeoVLN achieves strong performance under a single-RGB setting without additional navigation-specific external training data. More importantly, we construct controlled ablations that separate the effect of *multi-depth representation selection* from the effect of simply performing more geometry injections, and independently study the relationship between geometric-memory budget and navigation performance. Beyond aggregate benchmark scores, we further evaluate the resulting policy on spatially demanding simulation trajectories and extend the evaluation to a physical Unitree G1 humanoid, providing a complementary test beyond simulator-only metrics.

Our contributions are:

- **Geometric reasoning as selective allocation across representation depth.** We question the use of a single terminal geometry representation and introduce hierarchical GFM-VLM fusion that exposes multiple GFM depths to successive policy stages. Controlled comparisons against repeated terminal-feature injection isolate whether representational depth itself contributes to navigation reasoning.
- **Streaming geometric memory as navigation-conditioned selection across time.** We introduce a bounded GFM memory that determines which historical geometric states remain available for future reasoning according to instruction relevance, geometric confidence, and trajectory transitions, rather than treating all past states equally or relying only on recency.
- **Controlled empirical validation across benchmark and physical settings.** We evaluate AdaGeoVLN on R2R-CE and RxR-CE, conduct targeted ablations that separately interrogate the proposed depth- and time-selection hypotheses, and extend evaluation beyond simulation through deployment on a Unitree G1 humanoid. This combination examines both the source of the observed gains and their relevance beyond standard VLN evaluation.

II. RELATED WORK

a) Vision-Language Navigation. VLN was introduced by Anderson et al. [2] to study instruction following in previously unseen indoor environments. Early methods largely learned recurrent mappings from natural-language instructions and panoramic observations to navigation actions. Speaker-Follower models [3] improved data efficiency through synthetic instruction generation and pragmatic route reasoning, while EnvDrop [4] used environmental dropout and back-translation to improve generalization. PREVALENT [5] introduced large-scale VLN pretraining, and HAMT [6] explicitly encoded long-horizon panoramic histories with a hierarchical multimodal Transformer. DUET [7] further combined fine-grained local reasoning with a global topological representation. In parallel, VLN-CE [8] moved evaluation beyond oracle connectivity graphs to continuous

environments, where agents execute low-level actions with-out known environment topology or perfect localization. Recent systems increasingly use multimodal foundation models: NaVid and Uni-NaVid study video-centric navigation [9], [10], NaVILA couples vision-language planning with locomotion [11], StreamVLN explicitly models streaming context [12], and JanusVLN separates spatial-geometric and visual-semantic implicit memories [13]. Our focus is complementary: how a pretrained GFM itself should be consumed across internal depth and retained across navigation time.

b) Spatial and Geometric Representations for VLN.:

A complementary line of work introduces explicit spatial structure to overcome limitations of purely appearance-based reasoning. Structured Scene Memory [14] organizes visual and geometric observations into a structured external memory for long-range reasoning. GridMM [15] projects historical observations into a dynamically growing top-down grid and aggregates instruction-relevant features within spatial cells. Bird’s-Eye-View Scene Graph [16] combines local BEV representations with a global scene graph, while Volumetric Environment Representation [17] lifts multi-view observations into structured 3D cells for occupancy, room-layout, and object reasoning. These approaches demonstrate the value of geometry, but generally *construct* a task-specific spatial representation—for example, a topological graph, grid, BEV map, or voxel volume—and then expose the resulting representation to the policy. AdaGeoVLN asks a different question: given a pretrained geometry model that already develops intermediate representations throughout its processing hierarchy, should VLN consume only its terminal state, or should different geometric depths interact with different stages of navigation reasoning?

c) Geometry Foundation Models and Geometry-Augmented MLLMs.: Large-scale geometric representation learning has shifted from task-specific depth and reconstruction networks toward broadly transferable models. Depth Anything [18] demonstrates scalable monocular-depth pretraining, while DUST3R [19] reformulates multi-view reconstruction through learned point-map prediction and MAST3R [20] extends this paradigm toward geometry-grounded image matching. VGGT [1] further unifies camera estimation, depth, point-map prediction, and point tracking in a feed-forward Transformer. In multimodal reasoning, VG-LLM [21] integrates geometry priors from video into an MLLM, and SpatialStack [22] explores multi-level geometry for spatial reasoning. These works motivate the geometry-language interface used here. Our contribution is not the generic act of attaching VGGT to a VLM; rather, we study the role of the *GFM representation hierarchy* in sequential embodied decision-making and test whether preserving access to multiple depths matters for VLN.

d) Long-Horizon Memory and Streaming Navigation.:

Memory has remained central throughout VLN. HAMT [6] directly encodes past panoramic observations, while Structured Scene Memory [14], GridMM [15], and DUET [7] organize experience through explicit scene memories or maps. Iterative VLN [23] further shows that simply ex-

tending Transformer history is insufficient for persistent navigation, whereas explicit environmental memory can benefit from scene persistence. Recent foundation-model approaches increasingly compress or reuse neural context. StreamVLN [12] maintains compact streaming context, while JanusVLN [13] retains fixed-size spatial-geometric and visual-semantic KV memories using initial and sliding windows. AdaGeoVLN operates directly on the global-attention KV states of the GFM and asks which historical geometric states should remain available to *future geometric inference*. Retention is therefore conditioned on navigation utility—instruction relevance, GFM confidence, and trajectory-transition evidence—rather than recency alone.

e) Physical Embodiment of VLN.: Simulation abstracts away sensing and control effects that can become important on physical robots. NaVILA [11] connects high-level vision-language reasoning with learned locomotion on legged platforms, including humanoids. VLN-PE [24] systematically studies the simulator-to-physical gap across humanoid, quadruped, and wheeled embodiments, reporting degradation from restricted observations, visual variation, collisions, and locomotion constraints. Motivated by this gap, we complement benchmark evaluation with deployment on a Unitree G1 humanoid; the physical experiment is intended to assess transfer of the high-level navigation policy rather than contribute a new locomotion controller.

III. METHOD

A. Problem Formulation and Overview

At navigation step t , the agent receives an RGB observation I_t and a natural-language instruction $\mathcal{L}$, and predicts an action $a_t \in \mathcal{A}$ conditioned on the current observation and navigation history. We consider a geometry foundation model (GFM) $\mathcal{G}$ that processes the visual stream and exposes intermediate representations $\{G_t^m\}$ at multiple depths m , together with a vision-language navigation policy $\mathcal{P}$. The action policy can be written as

$$a_t \sim \pi_\theta(a_t \mid \mathcal{L}, I_{1:t}, \mathcal{M}_t), \quad (1)$$

where $\mathcal{M}_t$ denotes the retained geometry-side memory.

Rather than treating geometry as a single auxiliary feature, we study two orthogonal selection problems. The first concerns *representation depth*: given the hierarchy of intermediate states produced by $\mathcal{G}$, which geometric representations should interact with different stages of language-conditioned navigation reasoning? The second concerns *navigation time*: as geometric states accumulate along a trajectory, which historical states should remain available to future geometric inference under a bounded memory budget?

AdaGeoVLN addresses these questions through two coupled mechanisms. First, hierarchical GFM–VLM fusion exposes geometric representations from multiple GFM depths to successive stages of the navigation policy. Second, navigation-aware GFM memory selectively preserves historical global-attention KV states according to their utility for subsequent navigation. Thus, the framework performs

selective geometric reasoning along two axes: *which representation to use across model depth* and *which representation to retain across navigation time*. Figures 2 and 3 detail the two mechanisms.

B. Hierarchical Geometry Fusion for Navigation Reasoning

Recent geometry-augmented multimodal models show that latent features from a pretrained geometry encoder can complement conventional visual tokens for spatial reasoning [21], [22], [13]. We build on this general geometry-language interface while asking a question specific to sequential navigation: *should a navigation policy interact with geometry only through a terminal GFM representation, or should the GFM hierarchy remain accessible across successive policy stages?*

Following the multi-depth interface used in Spatial-Stack [22], we extract VGGT representations at depths $m \in \{11, 17, 23\}$ and align them with the first three language-decoder stages $\ell \in \{0, 1, 2\}$:

$$(G_t^{11}, G_t^{17}, G_t^{23}) \longrightarrow (H_t^0, H_t^1, H_t^2). \quad (2)$$

Importantly, AdaGeoVLN does not assign predefined semantic roles to these depths. Instead, the hierarchy is treated as an empirical hypothesis: latent states obtained at different stages of geometric processing may provide complementary information for navigation that is obscured when only one terminal state is repeatedly injected.

For each depth m , the VGGT representation is normalized and spatially aligned with the VLM image-token resolution:

$$Z_t^m = \text{Group}_{2 \times 2}(\text{RMSNorm}(G_t^m)). \quad (3)$$

VGGT provides a 2048-D geometry feature at each spatial location; grouping four neighboring patches yields an 8192-D token. A depth-specific adapter maps the grouped representation to the 2560-D language hidden space,

$$\Phi_m : \mathbb{R}^{8192} \rightarrow \mathbb{R}^{4096} \rightarrow \mathbb{R}^{2560}, \quad (4)$$

and a token-wise gate

$$\Gamma_m : \mathbb{R}^{8192} \rightarrow \mathbb{R}^{2048} \rightarrow \mathbb{R} \quad (5)$$

ends with a sigmoid. The resulting geometry update is

$$\Delta_t^m = s_m [\sigma(\Gamma_m(Z_t^m)) \odot \Phi_m(Z_t^m)], \quad (6)$$

where s_m is a learned scalar controlling the contribution of depth m . Geometry is added only to image-token positions through a residual update,

$$H_{t,\text{img}}^{\ell,+} = H_{t,\text{img}}^{\ell} + \Delta_t^m, \quad (m, \ell) \in \{(11, 0), (17, 1), (23, 2)\}. \quad (7)$$

Text-token states are not directly overwritten; subsequent decoder layers propagate the geometry-enriched visual context into language-conditioned action reasoning.

The purpose of this design is not merely to increase the number of geometry injections. We test this distinction explicitly by comparing the hierarchical configuration (G^{11}, G^{17}, G^{23}) against a controlled configuration that injects the same terminal representation G^{23} at all three policy stages (Sec. IV-C).

Hierarchical VGGT-VLM Fusion

Multi-depth hierarchical geometry injection

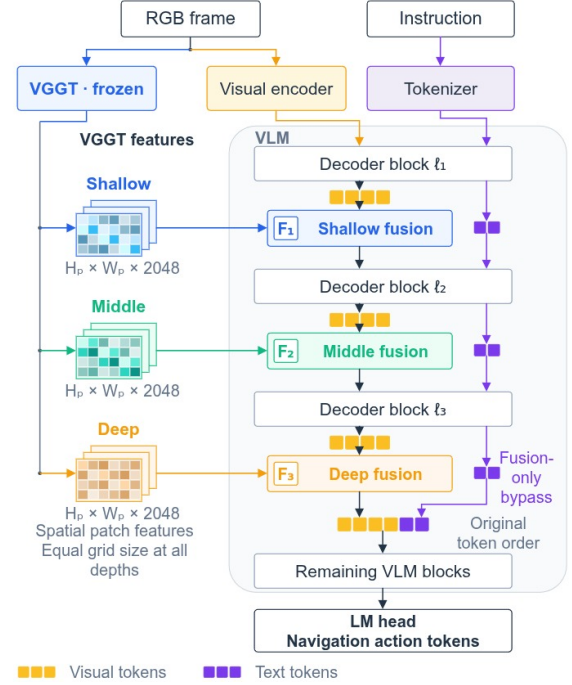


Fig. 2: **Hierarchical GFM-VLM fusion across representation depth.** VGGT representations from depths 11, 17, and 23 are coupled to successive Qwen decoder stages. Each branch applies RMSNorm, 2×2 spatial grouping, a depth-specific projection, a token-wise gate, and a learned scale before residual addition at image-token positions. The layer identities indicate relative depth; no fixed semantic role is assumed for a particular level.

C. Navigation-Aware GFM KV Retention

The second selection problem occurs over navigation time. In streaming inference, each VGGT global-attention layer can reuse key-value (KV) states produced by previous observations. Keeping the complete history preserves all geometric context, but its memory grows with trajectory length; a purely temporal window, in contrast, assumes that usefulness is determined by age. We instead formulate GFM memory as a *navigation-conditioned token-selection problem*: at every step, historical and newly produced VGGT states compete under a fixed memory budget according to their relevance to the current navigation task. The retained state is the *VGGT global-attention KV cache*, not the Qwen/VLM KV cache. A separate CPU-side buffer stores frame-aligned $G^{11/17/23}$ features required by the hierarchical fusion in Sec. III-B.

a) *Candidate memory.*: For global-attention layer ℓ , the current forward pass appends the KV states of frame t to the states retained from the previous step,

$$\mathcal{C}_t^\ell = M_{t-1}^\ell \cup (K_t^\ell, V_t^\ell). \quad (8)$$

The candidate cache is scored and pruned only *after* the current VGGT forward pass. Consequently, the retention decision at t controls the geometric context available at

Navigation-aware geometry KV eviction

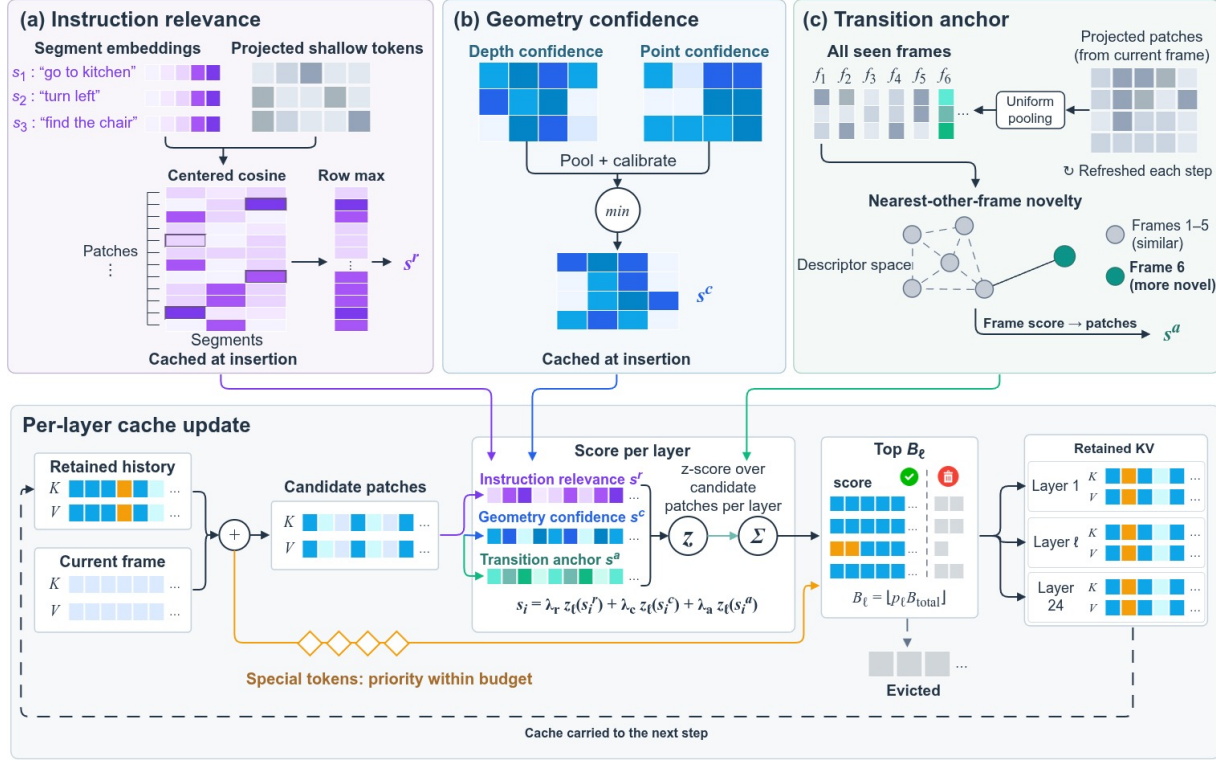


Fig. 3: **Navigation-aware GFM KV retention across navigation time.** Historical and current VGGT global-attention states form the candidate memory. Patch tokens are ranked using three complementary signals: instruction relevance, GFM confidence, and transition importance. Scores are normalized over the candidate patch population before each global-attention layer independently retains its highest-ranked states under the assigned budget. The resulting bounded KV memory is reused by VGGT at the next observation.

$t + 1$ and does not retroactively modify the current-frame representation.

b) Instruction-conditioned relevance.: A navigation instruction generally contains multiple ordered clauses whose relevance changes along a trajectory. We therefore decompose the instruction into action-oriented segments using punctuation, sequencing markers (e.g., “then”, “next”, and “after that”), and conjunctions that introduce a navigation verb, while preserving spatial phrases such as “next to.” For each segment j , we mean-pool its Qwen input-token embeddings and ℓ_2 -normalize the result to obtain a static episode-level embedding e_j .

Let $u_{t,p} \in \mathbb{R}^{2560}$ denote the current geometry token after alignment to the Qwen language space and before the depth-specific gate and scale. Its instruction relevance is the similarity to the most compatible instruction segment,

$$r_{t,p} = \max_j \frac{1 + \cos(u_{t,p}, e_j)}{2}. \quad (9)$$

The maximum over segments avoids forcing every retained geometric token to align with the complete instruction and instead measures whether it remains useful to at least one navigation sub-goal.

c) GFM confidence.: VGGT provides depth and point confidence together with its geometric predictions. Because the raw VGGT confidence is produced in $(1, \infty)$, we first

pool each confidence field onto the aligned token grid and monotonically map a raw value $\tilde{c}$ to $1 - 1/\tilde{c} \in [0, 1)$. The two confidence sources are then conservatively merged by their minimum,

$$c_{t,p} = \min(c_{t,p}^{\text{depth}}, c_{t,p}^{\text{point}}). \quad (10)$$

This term estimates the geometric reliability of each candidate patch rather than its task relevance.

d) Transition importance.: Instruction relevance and confidence are token-level quantities, whereas navigation also benefits from preserving observations around changes in viewpoint or scene structure. In the reported three-component profile, we construct a confidence-weighted descriptor of the current aligned geometry,

$$d_t = \text{norm} \left(\frac{\sum_p c_{t,p} u_{t,p}}{\sum_p c_{t,p} + \epsilon} \right). \quad (11)$$

Let $\mathcal{R}_4$ be the descriptors of the four preceding frames. We measure the current frame’s transition novelty as

$$\nu_t = \frac{1 - \text{mean}_{i \in \mathcal{R}_4} \cos(d_t, d_i)}{2}, \quad (12)$$

with $\nu_0 = 0$. The frame-level novelty is distributed over its patch tokens using their geometric reliability and instruction relevance,

$$a_{t,p} = \nu_t (0.5c_{t,p} + 0.5r_{t,p}). \quad (13)$$

Thus, transition importance does not reward motion alone: a novel frame contributes more strongly through patches that are geometrically reliable, instruction-relevant, or both.

e) Cache-wide token scoring.: For a candidate patch token originating from frame f and patch p , we retain three complementary quantities: confidence $c_{f,p}$, instruction relevance $r_{f,p}$, and transition importance $a_{f,p}$. Their raw scales differ, so immediately before eviction each component is standardized over the complete candidate *patch-token* population of layer ℓ . Denoting this layer- and step-specific standardization by $z_{t,\ell}(\cdot)$, the score is

$$S_{t,\ell}(f, p) = \lambda_c z_{t,\ell}(c_{f,p}) + \lambda_r z_{t,\ell}(r_{f,p}) + \lambda_a z_{t,\ell}(a_{f,p}). \quad (14)$$

We use fixed equal coefficients $\lambda_c = \lambda_r = \lambda_a$ for the reported experiments. They are a simple non-tuned combination, *not* an estimate of optimal relative importance. Our analysis in Sec. IV-D.0.b instead tests the scientific question relevant here: whether each constituent signal contributes useful information by removing one component at a time under the same budget. The implementation also exposes geometry-only and recency terms for controlled variants, but both have zero weight in the reported three-component profile.

f) Per-layer bounded retention.: Each of the 24 VGGT global-attention layers receives a fixed token budget

$$B_\ell = \lfloor \rho_\ell B_{\text{total}} \rfloor, \quad \sum_{\ell=1}^{24} \rho_\ell = 1, \quad (15)$$

and independently retains the highest-scoring candidates,

$$M_t^\ell = \text{TopK}(C_t^\ell, S_{t,\ell}, B_\ell). \quad (16)$$

The reported “900K” setting refers to the aggregate number of layer-token slots B_{total} across the 24 global-attention layers; it does not denote 900K unique scene tokens. VGGT’s special camera/register tokens are excluded from patch-score normalization and receive an explicit retention privilege above the best patch score, preventing the eviction rule from discarding the structural tokens required by the geometry backbone. Because each layer performs Top- K independently, different global-attention layers may retain different subsets of the historical geometric stream.

D. Training and Inference

The navigation policy uses Qwen3.5-4B as the VLM and a frozen VGGT-1B geometry backbone. Qwen receives the instruction and up to nine RGB observations; for longer trajectories, historical RGB frames are uniformly sampled while retaining the current observation. VGGT receives only the current RGB frame together with the retained historical GFM KV defined above. During supervised training, the hierarchical geometry interface is optimized jointly with the navigation policy while the GFM remains frozen. At inference time, the language head generates a textual action that is parsed into one of four Habitat navigation actions. All memory-retention variants use the same frozen hierarchical checkpoint and differ only in inference-time GFM retention, isolating temporal-memory effects from policy retraining.

IV. EXPERIMENTS

A. Experimental Setup

a) Benchmarks.: We evaluate on R2R-CE [2], [8] and RxR-CE [25] Val-Unseen. R2R-CE evaluates continuous-environment instruction following on Matterport3D scenes, while RxR-CE provides longer and linguistically more diverse trajectories. We report Navigation Error (NE, lower is better), Oracle Success (OS), Success Rate (SR), Success weighted by Path Length (SPL), and normalized Dynamic Time Warping (nDTW) where reported. The main comparison uses the evaluation protocol and metrics reported by the corresponding baseline papers.

b) Observation and training regime.: Our navigation policy receives a single egocentric RGB stream and language instruction; it does not consume panoramic observations, odometry, or simulator-provided depth at inference. The policy uses Qwen3.5-4B with a frozen VGGT-1B geometry encoder. The navigation checkpoint is trained on the standard R2R/RxR-CE training data used by our benchmark setting and does not use additional navigation-specific external data. All controlled geometry variants use the same navigation backbone and training mixture. Memory variants use the same frozen hierarchical checkpoint and differ only in inference-time GFM retention.

c) Memory measurement.: We separately report the memory occupied by retained VGGT global-attention KV and the end-to-end GPU allocation. Unless otherwise stated, memory-budget experiments are evaluated on R2R-CE Val-Unseen. Hardware comparisons report peak allocated memory on an RTX 6000 Pro; we additionally record whether configurations fit on an RTX 5090. The aggregate token budgets (e.g., 900K) count layer-token slots across all 24 VGGT global-attention layers.

B. Main Benchmark Results

Table I compares AdaGeoVLN with representative methods under different sensing and training regimes. We report panoramic, odometry/depth-based, and single-RGB methods separately because their observation assumptions differ substantially, and we additionally expose extra navigation supervision when it is quantified by the corresponding work. Within the single-RGB streaming setting, AdaGeoVLN achieves strong performance on both benchmarks without additional navigation-specific training data. On R2R-CE Val-Unseen, it obtains 55.3% SR and 50.0% SPL, improving over JanusVLN* while remaining competitive in NE and OS. On the more linguistically and trajectory-diverse RxR-CE benchmark, AdaGeoVLN reaches 53.7% SR and 62.0 nDTW, compared with 51.4% SR and 59.1 nDTW for JanusVLN*.

The improvement across both benchmarks suggests that the benefit of geometry-aware reasoning is not restricted to a single instruction distribution. Moreover, these results are obtained from a single RGB stream and without additional navigation training samples. Methods trained with substantially larger navigation corpora remain informative

TABLE I: **R2R-CE and RxR-CE Val-Unseen comparison.** Pano., Odo., Depth, and S.RGB denote panoramic, odometry, depth, and single-RGB observations. Extra Nav. Data counts additional navigation samples beyond the standard benchmark training sets when reported. Bold/underline mark best/second-best within the single-RGB streaming/VLM group.

Method	Observation				R2R-CE Val-Unseen				RxR-CE Val-Unseen				Training	
	Pano.	Odo.	Depth	S.RGB	NE↓	OS↑	SR↑	SPL↑	NE↓	SR↑	SPL↑	nDTW↑	Extra Nav.	Data
<i>Methods using panoramic/odometry/depth observations</i>														
HPN+DN [26]	✓	✓	✓	–	6.31	40.0	36.0	34.0	–	–	–	–	–	–
Sim2Sim [27]	✓	✓	✓	–	6.07	52.0	43.0	36.0	–	–	–	–	–	–
VLN ² BERT [28]	✓	✓	✓	–	5.74	53.0	44.0	39.0	8.98	27.0	22.6	46.7	–	–
Ego ² -Map [29]	✓	✓	✓	–	5.54	56.0	47.0	41.0	–	–	–	–	–	–
DreamWalker [30]	✓	✓	✓	–	5.53	59.0	49.0	44.0	–	–	–	–	–	–
InstructNav [31]	✓	–	–	–	6.89	–	31.0	24.0	–	–	–	–	–	–
COSMO [32]	✓	–	–	–	–	56.0	47.0	40.0	–	–	–	–	–	–
AO-Planner [33]	✓	–	✓	–	5.55	59.0	47.0	33.0	7.06	43.3	30.5	50.1	–	–
LAW [34]	–	✓	✓	✓	6.83	44.0	35.0	31.0	10.90	8.0	8.0	38.0	–	–
g3D-LF [35]	–	✓	✓	✓	5.70	59.5	47.2	34.6	–	–	–	–	–	–
Seq2Seq [8]	–	–	✓	✓	7.77	37.0	25.0	22.0	12.10	13.9	11.9	30.8	–	–
NaVid-4D [36]	–	–	✓	✓	5.99	55.7	43.8	37.1	–	–	–	–	–	–
NavMorph [37]	–	–	✓	✓	5.75	56.9	47.9	33.2	8.85	30.8	22.8	44.2	–	–
<i>Single-RGB streaming / VLM-based methods</i>														
NaVid [9]	–	–	–	✓	5.47	49.1	37.4	35.9	–	–	–	–	953K	–
Sim2Real [38]	–	–	–	✓	5.95	55.8	44.9	30.4	8.79	36.7	25.5	18.1	0K	–
StreamVLN [12]	–	–	–	✓	5.43	62.5	52.8	47.2	6.72	48.6	42.5	60.2	0K	–
Uni-NaVid [10]	–	–	–	✓	5.58	53.3	47.0	42.7	6.24	48.7	40.9	–	3577K	–
NaVILA* [11]	–	–	–	✓	5.37	57.6	49.7	45.5	–	–	–	–	12574K	–
NaVILA [11]	–	–	–	✓	<u>5.22</u>	62.5	<u>54.0</u>	49.0	6.77	49.3	44.0	58.8	13132K	–
JanusVLN* [13]	–	–	–	✓	5.17	58.0	52.8	<u>49.2</u>	6.46	<u>51.4</u>	44.3	<u>59.1</u>	0K	–
AdaGeoVLN	–	–	–	✓	<u>5.22</u>	<u>61.5</u>	55.3	50.0	5.54	53.7	44.3	62.0	0K	–

* follows the corresponding paper’s model-variant notation. NaVILA* excludes human-following data. Extra navigation data should be considered when comparing accuracy across training regimes.

TABLE II: **Controlled geometry study on R2R-CE Val-Unseen.** Deep×3 matches the three fusion locations while repeatedly using the same terminal representation.

Geometry	SR↑	SPL↑	OS↑	NE↓
None	46.5	42.3	56.0	5.19
Single Deep $G^{23} \rightarrow L_2$	<i>running</i>			
Deep×3 G^{23}	42.1	36.9	56.7	6.64
Hier. $G^{11/17/23}$	55.3	50.00	61.5	5.22

comparisons, but their accuracy should be interpreted jointly with the supervision reported in the final column of Table I.

C. Does Geometric Representation Depth Matter?

The first hypothesis asks whether multi-depth geometry itself matters, or whether the same benefit can be obtained simply by injecting a terminal representation multiple times. Table II therefore compares the hierarchical model with two controls: a geometry-free Qwen3.5 policy and a repeated-deep model that uses the same three fusion sites but injects G^{23} at each site.

Hierarchical fusion improves SR/SPL by 8.76/7.63 points over the matched geometry-free policy. More importantly, it exceeds the repeated-deep control by 13.16/13.07 points.

Because Deep×3 uses the same number and positions of fusion operations, the difference cannot be explained by simply increasing geometry injection frequency. Repeated terminal-feature injection is not sufficient to recover the gain, supporting the hypothesis that preserving access to representations from multiple points in the GFM hierarchy matters for navigation reasoning. We deliberately avoid assigning a fixed semantic meaning to G^{11} , G^{17} , or G^{23} ; the experiment establishes the value of multi-depth access, not a hand-specified interpretation of each level. The remaining single-deep $G^{23} \rightarrow L_2$ control will further isolate how much benefit is obtained from one terminal injection alone.

D. Navigation-Aware Geometric Memory

Our second hypothesis concerns which historical geometric states should remain available under a bounded memory budget. We study this from three complementary perspectives: the memory–performance trade-off, the contribution of each navigation signal, and a matched-capacity temporal baseline.

a) *Memory–performance trade-off.* Without navigation-aware pruning, the hierarchical model uses 5767.55 MB mean GFM-KV memory and reaches 55.25%

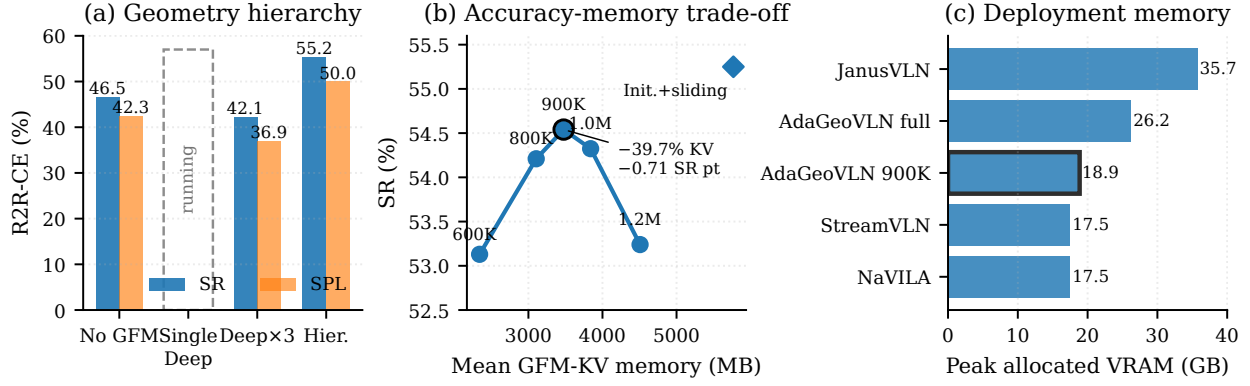


Fig. 4: **Controlled geometry and memory evidence.** (a) R2R-CE geometry controls compare no geometry, repeated terminal geometry, and hierarchical multi-depth geometry. (b) R2R-CE SR versus measured GFM-KV memory across retention budgets. The final 900K rerun reaches 54.54% SR with 39.7% lower mean GFM-KV memory than unpruned streaming geometry. (c) Peak allocated VRAM on RTX 6000 Pro.

TABLE III: **Leave-one-signal-out retention ablation at the paired 900K setting on R2R-CE.** Memory usage is effectively matched across rows.

Retention score	SR $\uparrow$	SPL $\uparrow$	NE $\downarrow$	KV MB $\downarrow$
Full	55.14	50.91	5.286	3475.92
w/o Transition	54.27	50.26	5.405	3475.96
w/o Confidence	53.62	49.62	5.481	3475.85
w/o Instruction	54.21	50.11	5.449	3475.76

SR / 49.95% SPL. Budgeted retention reduces GFM-KV memory across the 600K–1.2M range (Fig. 4b). The final 900K rerun reaches 54.54% SR and 50.37% SPL with 3475.68 MB mean GFM-KV memory, corresponding to a 39.7% reduction in mean GFM-KV memory. Mean allocated GPU memory falls from 24161.94 MB to 19275.87 MB (20.2%), while peak allocated memory decreases from 26954.89 MB to 19344.00 MB (28.2%). The sweep is not monotonic: retaining more states does not automatically improve navigation, suggesting that bounded history is not merely an approximation to “keep everything” but also changes which past geometry competes for attention.

b) Contribution of the retention signals.: We next test whether instruction relevance, geometric confidence, and transition importance each contribute to token selection. These ablations were collected against the same earlier full-900K reference run and therefore form an internally paired study; we do not mix them numerically with the later 900K rerun used for the memory-efficiency headline. Table III shows that removing any one signal decreases SR and SPL while leaving GFM-KV usage effectively unchanged. Removing confidence causes the largest SR drop in this set, but our claim is intentionally weaker: all three signals provide complementary evidence for deciding which geometry to preserve. The experiment supports the constituent signals, not the optimality of the equal coefficients in Eq. (14).

c) Matched-capacity temporal retention.: A critical control is whether navigation-aware selection outperforms a purely temporal policy given the *same* memory capacity. We therefore compare the 900K navigation-aware cache with a matched-capacity initial-plus-recent retention policy. [TBD:

Insert final matched-memory SR/SPL/NE and update the causal claim.] Until this result is frozen, we use the memory-budget sweep to support bounded-memory efficiency and avoid claiming that the navigation-aware scoring rule itself is superior to every recency policy.

E. Memory and Runtime Efficiency

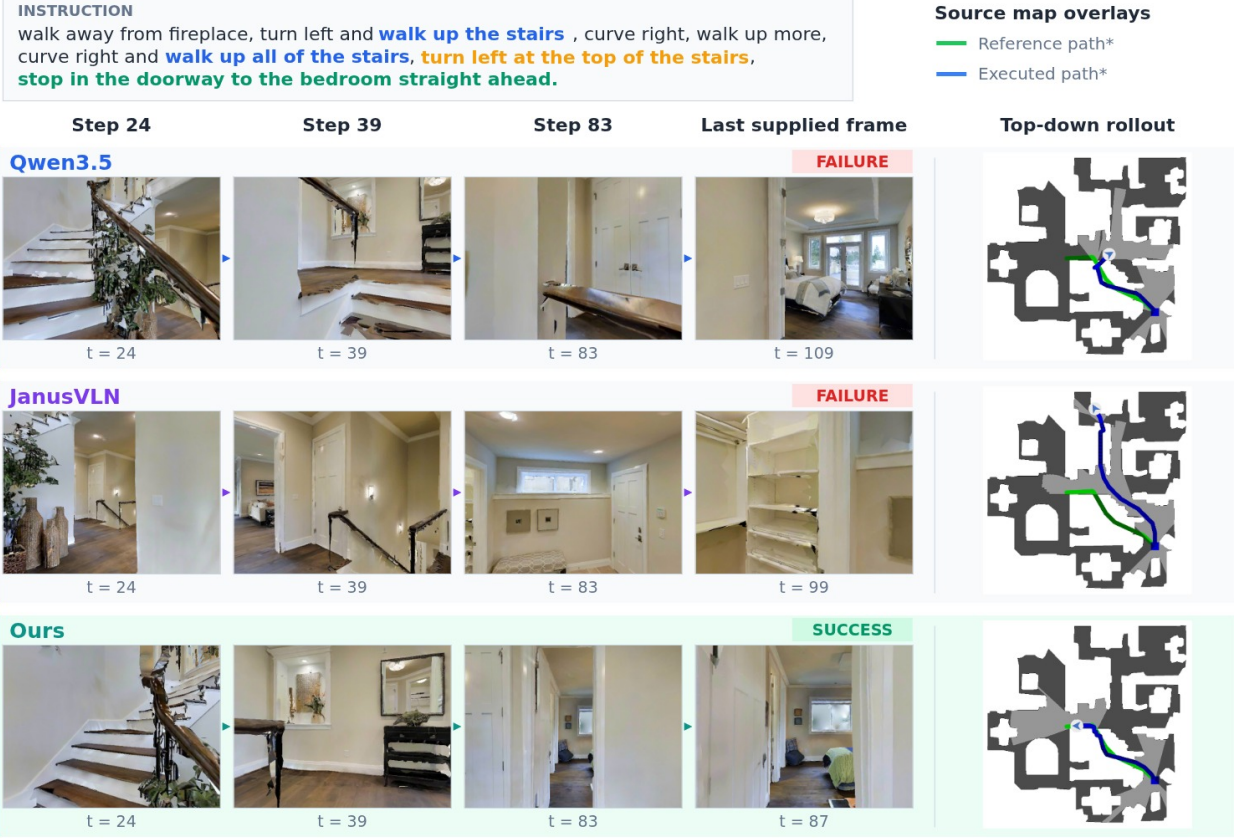
Figure 4(c) contextualizes the end-to-end cost of maintaining geometry-side state. On RTX 6000 Pro, the 900K configuration peaks at 19.31 GB allocated memory compared with 26.84 GB for the unpruned model, approaching the footprint of VLM-only streaming baselines. The same configuration also fits on RTX 5090. These hardware measurements support, rather than define, the temporal-memory claim: useful geometric history can be maintained under a bounded state budget.

F. Qualitative and Physical Humanoid Evaluation

a) Long-horizon spatial reasoning.: Fig. 5 compares rollouts on a challenging R2R-CE Val-Unseen episode requiring multi-level spatial reasoning. The instruction requires the agent to leave the initial fireplace region, enter and ascend a curved staircase, maintain the route through multiple direction changes, recognize the upper landing, and finally stop at a specified bedroom doorway. Both the Qwen3.5 policy and JanusVLN fail to complete the instructed route, whereas AdaGeoVLN reaches the target successfully. The top-down trajectory shows that AdaGeoVLN remains aligned with the reference route through the staircase and upper-floor transition. A single trajectory does not establish aggregate superiority; rather, it illustrates the kind of long-horizon spatial behavior targeted by preserving geometric information across both representation depth and navigation history.

b) Physical humanoid evaluation.: To complement simulator evaluation, we deploy the high-level VLN policy on a Unitree G1 humanoid in flat indoor environments. The robot receives an egocentric RGB stream and natural-language instruction, while an existing whole-body controller executes the navigation commands; locomotion control is not

Multi-level navigation: stair ascent, landing turn, and doorway stop



*Map colors are preserved from the supplied visualizations; steps are simulation indices, not elapsed time.

Fig. 5: **Qualitative comparison on a spatially demanding R2R-CE Val-Unseen episode (Episode 1121).** The instruction requires multi-level navigation involving a curved staircase, multiple direction changes, an upper-floor transition, and precise stopping at a bedroom doorway. Qwen3.5 and JanusVLN fail to complete the route, whereas AdaGeoVLN successfully reaches the target. Selected egocentric observations are shown together with the corresponding top-down trajectories; simulation step indices appear below each frame. Map colors follow the supplied trajectory visualizations.

a contribution of this work. The physical study is designed to test instruction grounding and spatial decision making under real sensing, viewpoint variation, and execution noise rather than to reproduce simulator dynamics exactly. [TBD: Insert final number of routes/trials, success criterion, success rate, and representative G1 sequence once the physical evaluation is frozen.]

V. DISCUSSION AND LIMITATIONS

The experiments support the central view that GFM integration for VLN should be analyzed along two distinct axes rather than treated as a generic geometry-augmentation problem. Across representation depth, the repeated-deep control is particularly informative: injecting the terminal G^{23} state at the same three policy locations performs substantially worse than exposing G^{11} , G^{17} , and G^{23} . This result rules out the simple explanation that the hierarchical model improves only because geometry is injected more often. At the same time, it does not imply a fixed semantic interpretation for each depth; understanding what information is uniquely represented at individual GFM layers remains an open direction.

Across navigation time, the memory-budget study shows that substantial geometry-side memory can be removed

without a commensurate drop in navigation performance, and the leave-one-signal-out study indicates that instruction relevance, geometric confidence, and transition importance each contribute to retention. However, two limitations should be stated explicitly. First, equal score coefficients are a fixed design choice rather than an optimized solution; the ablation establishes component utility, not optimal weighting. Second, the matched-capacity recency control is required before attributing an advantage specifically to the navigation-aware selection rule rather than to bounded retention in general.

The current study also evaluates one GFM/VLM pair and a fixed set of source depths inherited from the SpatialStack-style geometry interface. Broader backbone and layer-selection studies would be needed to establish whether the same depth pattern transfers across geometry and language models. Finally, physical Unitree G1 evaluation is intended as a test of high-level policy transfer under embodied sensing and control, not as evidence for a new locomotion method. Failure cases from the physical study should therefore be interpreted jointly with perception, navigation, and controller limitations.

VI. CONCLUSION

We introduced AdaGeoVLN, a streaming VLN framework that formulates geometry use as two coupled selection problems: which GFM representations should be exposed across policy depth, and which historical GFM states should be preserved across navigation time. Hierarchical fusion substantially outperforms both a geometry-free policy and a repeated-terminal-feature control, providing controlled evidence that multi-depth access matters beyond simply injecting geometry more often. Navigation-aware retention provides a complementary mechanism for bounding geometry-side history using task relevance, geometric reliability, and trajectory transitions, while preserving competitive navigation performance across a broad memory range. Together with strong R2R-CE and RxR-CE results, qualitative long-horizon rollouts, and physical humanoid evaluation, these findings support a broader view of geometry-aware embodied reasoning: the value of a geometry foundation model depends not only on what geometry it predicts, but on how its internal representations are allocated across model depth and navigation history.

REFERENCES

- [1] J. Wang, M. Chen, N. Karaev, A. Vedaldi, C. Rupprecht, D. Novotny, “Vggt: Visual geometry grounded transformer,” 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 5294–5306, 2025.
- [2] P. Anderson, Q. Wu, D. Teney, et al., “Vision-and-language navigation: Interpreting visually-grounded navigation instructions in real environments,” Proceedings of the IEEE conference on computer vision and pattern recognition, pp. 3674–3683, 2018.
- [3] D. Fried, R. Hu, V. Cirik, et al., “Speaker-Follower Models for Vision-and-Language Navigation,” Advances in Neural Information Processing Systems, 2018.
- [4] H. Tan, L. Yu, M. Bansal, “Learning to Navigate Unseen Environments: Back Translation with Environmental Dropout,” Proceedings of NAACL-HLT, pp. 2610–2621, 2019.
- [5] W. Hao, C. Li, X. Li, L. Carin, J. Gao, “Towards Learning a Generic Agent for Vision-and-Language Navigation via Pre-Training,” Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 13137–13146, 2020.
- [6] S. Chen, P. L. Guhur, C. Schmid, I. Laptev, “History Aware Multi-modal Transformer for Vision-and-Language Navigation,” Advances in Neural Information Processing Systems, 2021.
- [7] S. Chen, P. L. Guhur, M. Tapaswi, C. Schmid, I. Laptev, “Think Global, Act Local: Dual-Scale Graph Transformer for Vision-and-Language Navigation,” Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 16537–16547, 2022.
- [8] J. Krantz, E. Wijmans, A. Majumdar, D. Batra, S. Lee, “Beyond the nav-graph: Vision-and-language navigation in continuous environments,” European Conference on Computer Vision, pp. 104–120, 2020.
- [9] J. Zhang, K. Wang, R. Xu, et al., “Navid: Video-based vlm plans the next step for vision-and-language navigation,” arXiv preprint arXiv:2402.15852, 2024.
- [10] J. Zhang, K. Wang, S. Wang, et al., “Uni-navid: A video-based vision-language-action model for unifying embodied navigation tasks,” arXiv preprint arXiv:2412.06224, 2024.
- [11] A. C. Cheng, Y. Ji, Z. Yang, et al., “Navila: Legged robot vision-language-action model for navigation,” arXiv preprint arXiv:2412.04453, 2024.
- [12] M. Wei, C. Wan, X. Yu, et al., “Streamvln: Streaming vision-and-language navigation via slowfast context modeling,” arXiv preprint arXiv:2507.05240, 2025.
- [13] S. Zeng, D. Qi, X. Chang, et al., “Janusvln: Decoupling semantics and spatiality with dual implicit memory for vision-language navigation,” International Conference on Learning Representations, pp. 33001–33026, 2026.
- [14] H. Wang, W. Wang, W. Liang, C. Xiong, J. Shen, “Structured Scene Memory for Vision-Language Navigation,” Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 8455–8464, 2021.
- [15] Z. Wang, X. Li, J. Yang, Y. Liu, S. Jiang, “Gridmm: Grid memory map for vision-and-language navigation,” 2023 IEEE/CVF International Conference on Computer Vision (ICCV), pp. 15579–15590, 2023.
- [16] R. Liu, X. Wang, W. Wang, Y. Yang, “Bird’s-Eye-View Scene Graph for Vision-Language Navigation,” Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 10968–10980, 2023.
- [17] R. Liu, W. Wang, Y. Yang, “Volumetric Environment Representation for Vision-Language Navigation,” Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 16317–16328, 2024.
- [18] L. Yang, B. Kang, Z. Huang, X. Xu, J. Feng, H. Zhao, “Depth Anything: Unleashing the Power of Large-Scale Unlabeled Data,” Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 10371–10381, 2024.
- [19] S. Wang, V. Leroy, Y. Cabon, B. Chidlovskii, J. Revaud, “DUST3R: Geometric 3D Vision Made Easy,” Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 20697–20709, 2024.
- [20] V. Leroy, Y. Cabon, J. Revaud, “Grounding Image Matching in 3D with MAST3R,” Proceedings of the European Conference on Computer Vision, pp. 71–91, 2024.
- [21] D. Zheng, S. Huang, Y. Li, L. Wang, “Learning from Videos for 3D World: Enhancing MLLMs with 3D Vision Geometry Priors,” arXiv preprint arXiv:2505.24625, 2025.
- [22] J. Zhang, S. Zhou, B. Liu, A. Kadambi, Z. Fan, “Spatialstack: Layered geometry-language fusion for 3d vlm spatial reasoning,” arXiv preprint arXiv:2603.27437, 2026.
- [23] J. Krantz, S. Banerjee, W. Zhu, et al., “Iterative Vision-and-Language Navigation,” Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 14921–14930, 2023.
- [24] L. Wang, X. Xia, H. Zhao, et al., “Rethinking the Embodied Gap in Vision-and-Language Navigation: A Holistic Study of Physical and Visual Disparities,” Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 9455–9465, 2025.
- [25] A. Ku, P. Anderson, R. Patel, E. Ie, J. Baldridge, “Room-across-room: Multilingual vision-and-language navigation with dense spatiotemporal grounding,” Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), pp. 4392–4412, 2020.
- [26] J. Krantz, A. Gokaslan, D. Batra, S. Lee, O. Maksymets, “Waypoint models for instruction-guided navigation in continuous environments,” 2021 IEEE/CVF International Conference on Computer Vision (ICCV), pp. 15142–15151, 2021.
- [27] J. Krantz, S. Lee, “Sim-2-sim transfer for vision-and-language navigation in continuous environments,” European conference on computer vision, pp. 588–603, 2022.
- [28] Y. Hong, Z. Wang, Q. Wu, S. Gould, “Bridging the gap between learning in discrete and continuous environments for vision-and-language navigation,” 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 15418–15428, 2022.
- [29] Y. Hong, Y. Zhou, R. Zhang, et al., “Learning navigational visual representations with semantic map supervision,” 2023 IEEE/CVF International Conference on Computer Vision (ICCV), pp. 3032–3044, 2023.
- [30] H. Wang, W. Liang, L. Van Gool, W. Wang, “Dreamwalker: Mental planning for continuous vision-language navigation,” 2023 IEEE/CVF International Conference on Computer Vision (ICCV), pp. 10839–10849, 2023.
- [31] Y. Long, W. Cai, H. Wang, G. Zhan, H. Dong, “Instructnav: Zero-shot system for generic instruction navigation in unexplored environment,” arXiv preprint arXiv:2406.04882, 2024.
- [32] S. Zhang, Y. Qiao, Q. Wang, et al., “Cosmo: Combination of selective memorization for low-cost vision-and-language navigation,” 2025 IEEE/CVF International Conference on Computer Vision (ICCV), pp. 5511–5522, 2025.
- [33] J. Chen, B. Lin, X. Liu, L. Ma, X. Liang, K. Y. K. Wong, “Affordances-oriented planning using foundation models for continuous vision-language navigation,” Proceedings of the AAAI Conference on Artificial Intelligence, pp. 23568–23576, 2025.
- [34] S. Raychaudhuri, S. Wani, S. Patel, U. Jain, A. Chang, “Language-aligned waypoint (law) supervision for vision-and-language navigation

in continuous environments,” Proceedings of the 2021 conference on empirical methods in natural language processing, pp. 4018–4028, 2021.

- [35] Z. Wang, G. H. Lee, “g3d-1f: Generalizable 3d-language feature fields for embodied tasks,” 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 14191–14202, 2025.
- [36] H. Liu, W. Wan, X. Yu, et al., “Na vid-4d: Unleashing spatial intelligence in egocentric rgb-d videos for vision-and-language navigation,” 2025 IEEE International Conference on Robotics and Automation (ICRA), pp. 10607–10615, 2025.
- [37] X. Yao, J. Gao, C. Xu, “Navmorph: A self-evolving world model for vision-and-language navigation in continuous environments,” 2025 IEEE/CVF International Conference on Computer Vision (ICCV), pp. 5536–5546, 2025.
- [38] Z. Wang, X. Li, J. Yang, Y. Liu, S. Jiang, “Sim-to-real transfer via 3d feature fields for vision-and-language navigation,” arXiv preprint arXiv:2406.09798, 2024.